

Gigapower — EV-Battery Site-Selection Framework

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Executive Summary

Australia and **Canada** emerge as the unequivocal frontrunners —an expedited 60-day diligence window could lock in 15 GWh of capacity for $\leq$ US \$500 M while cutting supply-risk by roughly 130% compared to global average.



Undisputed Leadership

Gigafactory Attractiveness Index places **Australia #1** and **Canada #2**, both more than **1 σ** above the global mean on mineral depth and governance solidity.



Recommendation Resilience

“Risk-heavy” and **“Cost-heavy”** re-weightings retain 4 / 5 of today’s Top-5, confirming that the ranking is structurally robust to board-level trade-offs.



Clock-speed Advantage

A 60-day diligence sprint can advance two shortlisted sites to feasibility by **Q-next + 2 months**, preserving the 2029 start-of-production milestone.

Board seeks a de-risked **15 GWh cell supply** by 2029 within a **US \$500 M equity** envelope.



Capacity

15 GWH

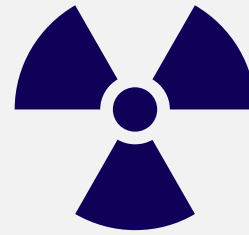
option to expand
to 50 GWh



Capital

US\$500 M

+ up to 30%
incentives

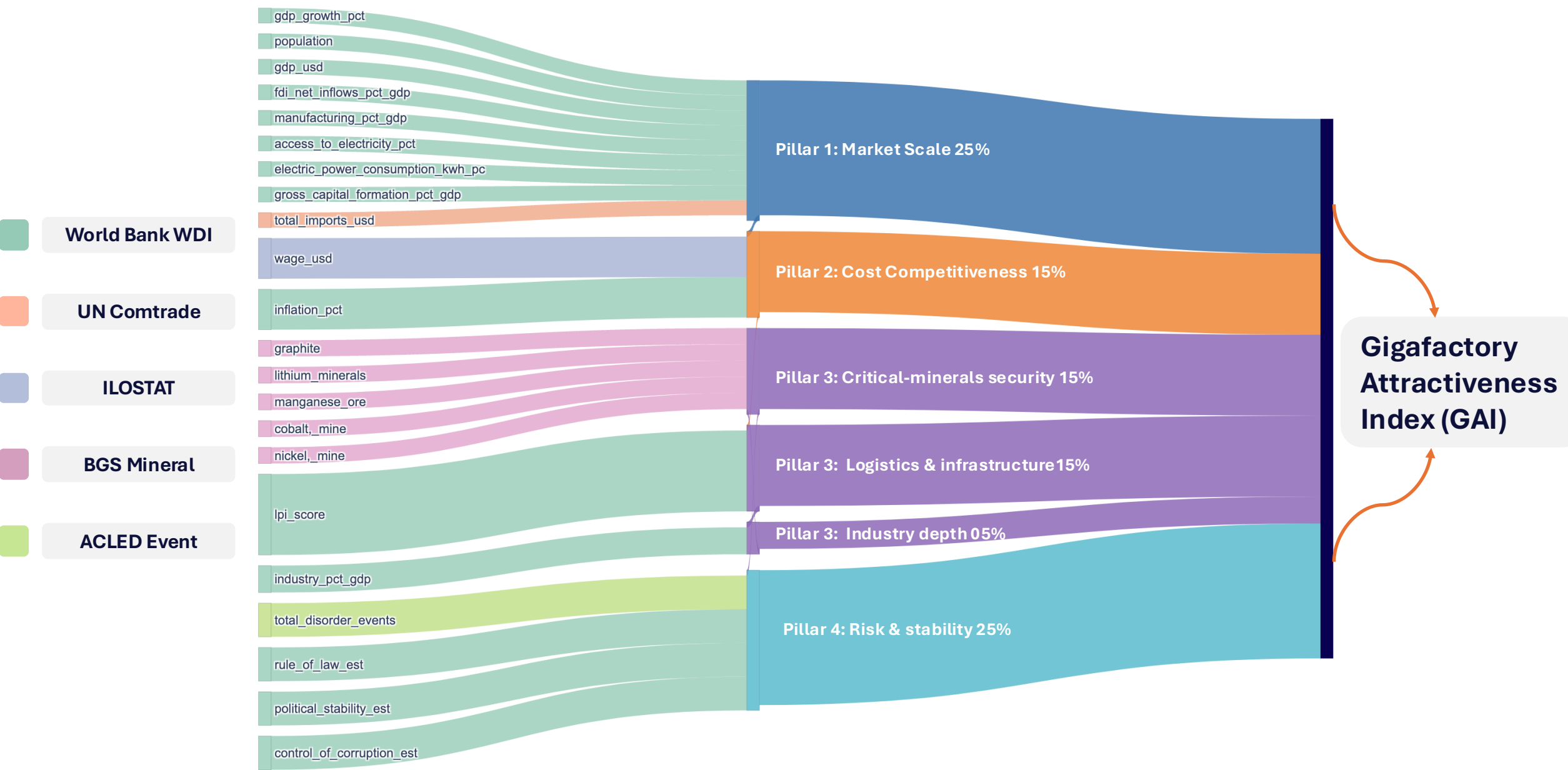


Risk

Above Median

composite
governance index

Five global datasets fused into an 80-country panel (2010-23) to form GAI.



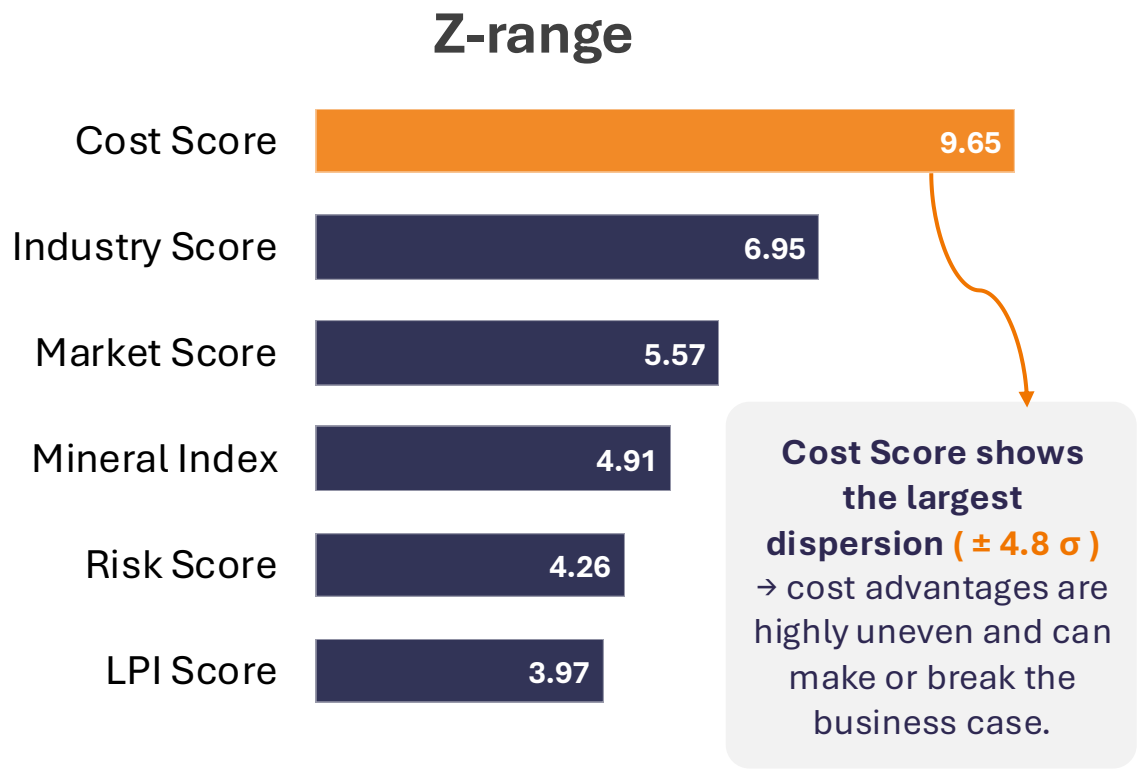
0 % missing after cleaning; variance normalized (z-score) across pillars.



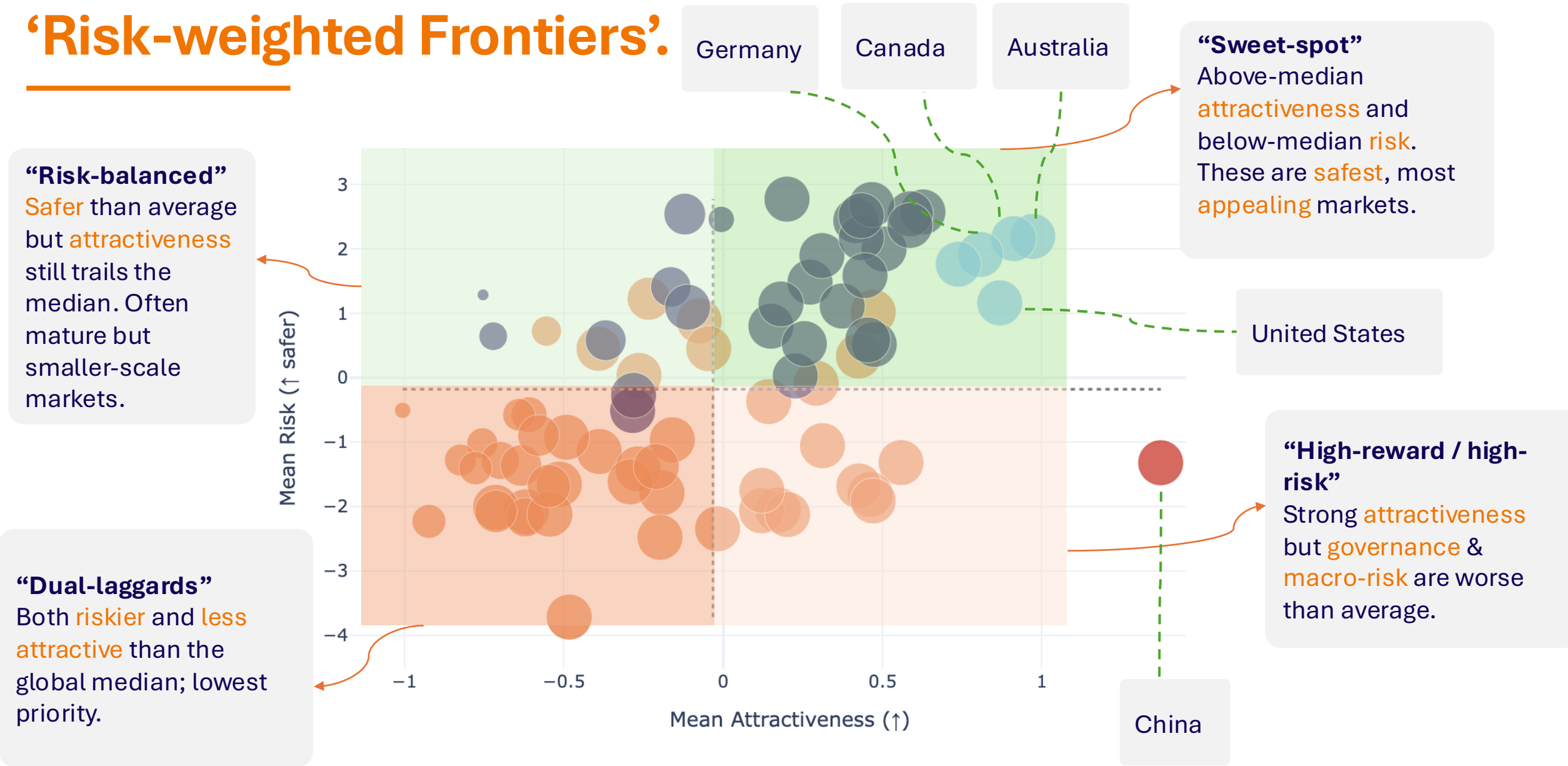
Data Completeness

Missing Values Scan

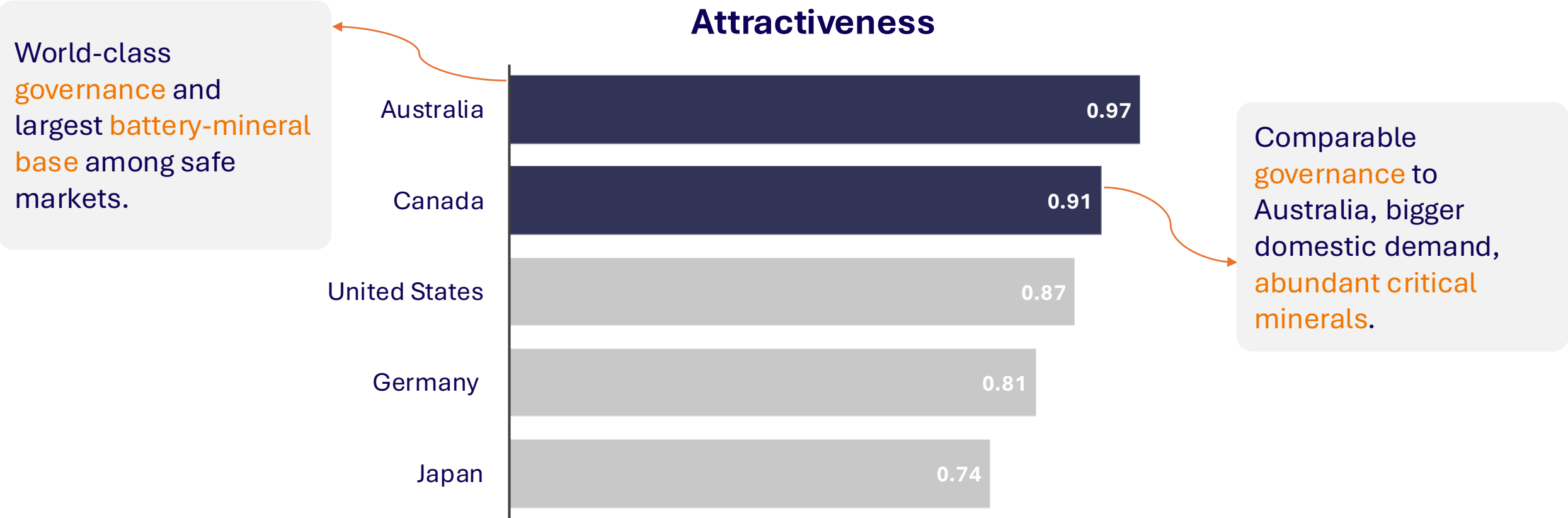
0 0 0 0 0 0



2 × 2 clustering shows ‘Safe Mature Hubs’ vs. ‘Risk-weighted Frontiers’.



Gigafactory Attractiveness Index isolates five “no-regret” markets.



Australia + Canada slashes supply-risk by 1.3 standard deviations but with same cost headwind.



Supply-risk ↓ 1.3 σ

Australia & Canada sit 1.3 standard deviations safer than the global average (robust legal systems, low conflict exposure).



Cost ↑ 1.3 σ

Labor + logistics run ~1.3 σ above the mean; the premium is the price of stability and market depth.

Cost (vs Global Average)

1.3 σ

-1.3 σ

Risk (vs Global Average)

Safe-Hub markets unlock an outsized **Mineral Security** advantage—nearly **3× richer** in battery-grade metals—while other pillars stay broadly in line.



Mineral Security
+ 2.9 σ

Australia + Canada sit 2.9 σ above the global mean on the **Mineral Index**, giving the venture access to ~3 × more battery-grade lithium, nickel, and cobalt per capita than a typical market.



Supply-chain efficiency
+1.2 σ

Their average Logistics Performance Index (LPI) score is 1.2 σ higher than the world baseline, signaling faster **customs clearance**, **denser freight networks** and **lower port-to-plant** transit times.

AU + CA vs Global Average



Appendix

Full-access links



Main Repo



Notebook 01



Notebook 02

